

Seminar Report (Literature Survey)

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ViMo: Generating Motions from Casual Videos (2024)

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1 Datasets Used

The paper utilizes several datasets for training and evaluation:

1.1 Primary Training Dataset

- **AIST++**: Contains 1,408 high-quality dance motions with music from 10 genres, with 3D motions and 2D videos from 8 camera angles.
- **Training Split**: 8 dance genres for training, **break** and **middle hip-hop** for testing.

1.2 Generated Dataset

- **Chinese Classic Dance Dataset**:
 - 52 solo-dance videos from Chinese annual dancing competition
 - Processed with AlphaPose for 2D pose extraction
 - Generated 750 3D motion clips (63 minutes total) using ViMo
 - Used for music-to-motion generation and style transfer tasks

1.3 Other Referenced Datasets

- Human3.6M
- AMASS
- KIT Motion-Language
- Finedance

2 Methodology

ViMo addresses the novel problem of **video-to-motion generation** from *casual videos* - videos characterized by complex camera movements, montages, occlusions, and varying perspectives. Given an input video $\mathcal{V}$ containing a single person performing actions, the goal is to generate multiple plausible 3D human motions $\mathbf{M}$ that are semantically aligned with the video content in terms of pose shape and rhythm, without requiring explicit camera parameter estimation.

2.1 Technical Framework Overview

Input Processing Pipeline

- **2D Pose Extraction:** Using pose estimators like AlphaPose or OpenPose to extract 2D skeleton sequences
- **Input Representation:** $p \in \mathbb{R}^{S \times J_{2d} \times C_{in}}$ where:
 - $S = 150$ (sequence length, 5 seconds at 30 FPS)
 - $J_{2d} = 17$ (COCO format joints)
 - $C_{in} = 3$ (x, y coordinates + confidence score)
- **Preprocessing:** Missing/low-confidence joints are filtered, interpolation applied for continuity

Output Motion Representation

The target 3D motion $\mathbf{M}$ is represented per frame as:

$$\mathbf{m} \in \mathbb{R}^{S \times (24 \times 6 + 4 + 3)} \quad (12.1)$$

Breaking down the 151-dimensional vector per frame:

- **Joint Rotations (144D):** 24 joints in 6D rotation representation
- **Foot Contact Labels (4D):** Binary indicators for [left foot, right foot, left toe, right toe]
- **Root Position (3D):** Global (x, y, z) translation of pelvis

2.2 Diffusion Model Foundation

Forward Noising Process

The diffusion process follows a Markov chain, progressively adding Gaussian noise:

$$q(m_t^{1:S} | m_{t-1}^{1:S}) = \mathcal{N}(\sqrt{\alpha_t} m_{t-1}^{1:S}, (1 - \alpha_t)I) \quad (12.2)$$

The closed-form expression for any step t given clean data m_0 is:

$$m_t = \sqrt{\bar{\alpha}_t} m_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon, \quad \epsilon \sim \mathcal{N}(0, I) \quad (12.3)$$

where $\bar{\alpha}_t = \prod_{s=1}^t \alpha_s$.

Reverse Denoising Process

The denoising network directly predicts the clean motion:

$$\hat{m}_0 = D(m_t, t, p) \quad (12.4)$$

The training objective uses the simple loss:

$$\mathcal{L}_{\text{simple}} = \mathbb{E}_{m \sim q(\mathbf{M}|p), t \sim [1, N]} [\|m_0 - D(m_t, t, p)\|_2^2] \quad (12.5)$$

Analytical Posterior Distribution

The true posterior $q(m_{t-1} | m_t, m_0)$ is Gaussian with known parameters:

$$\hat{\mu}_t(m_t, m_0) = \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1 - \bar{\alpha}_t}m_0 + \frac{\sqrt{\bar{\alpha}_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t}m_t \quad (12.6)$$

$$\tilde{\beta}_t = \frac{1 - \bar{\alpha}_{t-1}}{1 - \bar{\alpha}_t}\beta_t, \quad \text{where } \beta_t = 1 - \alpha_t \quad (12.7)$$

2.3 Network Architecture Details

Denoiser Network $D(m_t, t, p)$

- **Input:** Noisy motion m_t , diffusion timestep t , 2D pose sequence p
- **Architecture:** Transformer-based with three denoising modules
- **Components:**
 - Self-attention blocks for temporal coherence
 - FiLM conditioning blocks for feature-wise modulation
 - Cross-attention and MLP layers

FiLM Conditioning Mechanism

Feature-wise Linear Modulation dynamically scales and shifts features based on conditioning:

$$\mathbf{h}' = \gamma(\mathbf{c}) \odot \mathbf{h} + \beta(\mathbf{c}) \quad (12.8)$$

where:

- $\mathbf{h}$: Intermediate features of shape $n \times d$ (n frames, d hidden dimension)
- $\mathbf{c} = [\text{embed}(t); \text{encode}(p)]$: Concatenated timestep and pose embeddings
- γ, β : Functions mapping condition $\mathbf{c}$ to modulation parameters

Classifier-Free Guidance

During training, conditioning is randomly dropped 25% of the time. During sampling:

$$\hat{m}_0 = (1 + w)D(m_t, t, p) - wD(m_t, t, \phi) \quad (12.9)$$

where w controls guidance strength, balancing fidelity and diversity.

2.4 Motion Representation Details

6D Rotation Representation

Instead of problematic Euler angles or constrained quaternions, ViMo uses the continuous 6D representation:

$$\mathbf{R} = [\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_1 \times \mathbf{a}_2] \quad (12.10)$$

where $\mathbf{a}_1, \mathbf{a}_2$ are the first two columns of the rotation matrix. This ensures:

- Continuity for neural network learning
- No gimbal lock issues
- Easy reconstruction of orthonormal rotation matrix

Foot Contact Label Computation

Foot contact labels are computed self-supervised from foot velocities:

$$f_i = \mathbb{I}(\|\mathbf{V}_{\text{foot}}\|_2 < \epsilon) \quad (12.11)$$

where $\mathbf{V}_{\text{foot}}$ is the velocity of foot joints and ϵ is a threshold.

2.5 Loss Functions Formulation

The complete training objective combines diffusion and auxiliary losses:

$$\mathcal{L} = \mathcal{L}_{\text{simple}} + \lambda_1 \mathcal{L}_{\text{joints}} + \lambda_2 \mathcal{L}_{\text{vel}} + \lambda_3 \mathcal{L}_{\text{foot}} \quad (12.12)$$

Joints Loss (3D Position Error)

$$\mathcal{L}_{\text{joints}} = \frac{1}{S} \sum_{i=1}^S \left\| FK(R_0^i) - FK(\hat{R}_0^i) \right\|_2^2 \quad (12.13)$$

The Forward Kinematics function $FK(\cdot)$ converts joint rotations to 3D positions:

$$\mathbf{p}_j = \mathbf{p}_{\text{par}(j)} + \mathbf{R}_{\text{par}(j)} \cdot \mathbf{b}_j \quad (12.14)$$

where $\mathbf{b}_j$ is the bone vector in parent's local coordinates.

Velocity Loss (Temporal Smoothness)

$$\mathcal{L}_{\text{vel}} = \frac{1}{S-1} \sum_{i=1}^{S-1} \left\| (R_0^{i+1} - R_0^i) - (\hat{R}_0^{i+1} - \hat{R}_0^i) \right\|_2^2 \quad (12.15)$$

This ensures correct angular velocity and eliminates jitter by enforcing smooth transitions.

Foot Contact Loss (Physical Plausibility)

$$\mathcal{L}_{\text{foot}} = \frac{1}{S-1} \sum_{i=1}^{S-1} \left\| \left(FK \left(\hat{R}_0^{i+1} \right) - FK \left(\hat{R}_0^i \right) \right) \cdot \hat{f}_i \right\|_2^2 \quad (12.16)$$

Key aspects:

- Uses model’s own predicted contact labels $\hat{f}_i$
- Masks velocities of joints predicted to be in contact
- Creates self-supervised learning of foot-ground interaction

The masking operation:

$$\mathbf{V}_{\text{masked}}^i = \mathbf{m}_i \odot \mathbf{V}^i \quad (12.17)$$

where $\mathbf{m}_i$ is the contact mask broadcast across spatial dimensions.

2.6 Training and Sampling Procedure

Training Details

- **Sequence Length:** 150 frames (5 seconds at 30 FPS)
- **Diffusion Steps:** $T = 1000$
- **Optimizer:** ADAN with learning rate 1×10^{-4}
- **Sampling Acceleration:** DDIM with 50 steps
- **Conditioning:** Multi-view 2D poses from different camera angles

Sampling Process

At each denoising step:

1. Network predicts $\hat{m}_0 = D(m_t, t, p)$
2. Compute posterior mean using analytical formula:

$$\hat{\mu}_t = \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1 - \bar{\alpha}_t} \hat{m}_0 + \frac{\sqrt{\bar{\alpha}_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t} m_t \quad (12.18)$$

3. Sample $m_{t-1} \sim \mathcal{N}(\hat{\mu}_t, \tilde{\beta}_t I)$
4. Repeat until $t = 0$ for final motion $\hat{m}_0$

2.7 Advanced Applications Implementation

Zero-Convolution for Few-Shot Stylization

For adapting pre-trained models to new styles without catastrophic forgetting:

$$W_{\text{effective}} = W_{\text{pretrained}} + W_{\text{new}} \quad (12.19)$$

where W_{new} is initialized to zeros, ensuring:

- Initial behavior identical to pre-trained model
- Gradual adaptation to target style
- Preservation of music-dance alignment knowledge

Architecture uses residual connection:

$$\text{Output} = f_{\text{frozen}}(\text{input}) + f_{\text{zero-conv}}(\text{input}) \quad (12.20)$$

Video-Guided Motion Completion

For motion inpainting and editing tasks:

$$\hat{\mathbf{m}}_{t-1} = \mathbf{b} \odot \mathbf{q}(c_{\text{const}}, t-1) + (1 - \mathbf{b}) \odot \hat{\mathbf{m}}_{t-1} \quad (12.21)$$

where:

- $\mathbf{b}$: Binary mask indicating known regions
- c_{const} : Constraint motion provided by user
- Enables style transfer and missing motion completion

2.8 Key Technical Innovations

Robustness to Camera Complexity

- No explicit camera parameter estimation required
- Multi-view 2D pose conditioning provides view invariance
- Diffusion process handles ambiguity in 2D-to-3D lifting

Self-Supervised Physical Constraints

- Foot contact labels learned jointly with motion generation
- No manual annotation or fixed thresholds required
- Model discovers physically plausible contact patterns

Structured Motion Representation

- Combined rotation, translation, and contact information
- 6D rotation ensures continuity and avoids singularities
- Hierarchical loss functions for different motion aspects

This comprehensive methodology enables ViMo to handle the challenges of casual videos while generating diverse, physically plausible 3D motions that maintain semantic alignment with input video content.

2.9 Applications

Dataset Construction

ViMo generates large-scale motion datasets from video collections, demonstrated with Chinese classic dance dataset (750 motions from 52 videos).

Few-Shot Dancing Stylization

- Uses **zero-convolution blocks** for parameter-efficient fine-tuning
- Adapts pre-trained EDGE model to new styles
- Mathematical formulation: $W_{\text{effective}} = W_{\text{pretrained}} + W_{\text{new}}$ with W_{new} initialized to zeros

Video-Guided Motion Completion

Implements motion inpainting using masked denoising:

$$\hat{m}_{t-1} = b \odot q(c_{\text{const}}, t - 1) + (1 - b) \odot \hat{m}_{t-1} \quad (12.22)$$

where b is a binary mask for known regions.

3 Findings and Discussions

3.1 Quantitative Results

- **Superior to MotionBERT:** Lower FID scores (15.70 vs 19.22) and higher diversity
- **User Study:** 76.05% win rate vs MotionBERT’s 23.95%
- **Stylization:** Zero-convolution approach outperforms direct fine-tuning in user preference (56.27-65.89% win rates)

3.2 Qualitative Advantages

- Handles complex camera movements and occlusions better than traditional methods
- Generates smoother, more plausible motions without jittering
- Successfully transfers styles between different dance genres

3.3 Key Insights

- **Robustness:** Works on challenging casual videos where MotionBERT fails
- **Scalability:** Can amplify limited video resources into large motion datasets
- **Flexibility:** Enables three important applications beyond basic generation

4 Conclusion

ViMo represents a significant advancement in video-to-motion generation by:

- Leveraging abundant casual video resources beyond constrained MoCap data
- Introducing a robust diffusion-based framework that handles camera complexity
- Demonstrating practical applications in dataset construction, stylization, and motion completion
- Opening new directions for semantic-aware, scene-aware motion generation

The work successfully bridges the gap between casual video content and high-quality 3D motion generation, with potential impacts across animation, gaming, and virtual reality applications.

4.1 Improvements Made

- **Novel Problem Formulation:** First to address motion generation from casual videos with complex camera movements
- **Diffusion-based Approach:** Avoids explicit camera estimation, handles ambiguity better than deterministic methods
- **Self-supervised Foot Contact:** Eliminates need for manual labeling while preventing foot sliding
- **Multi-application Framework:** Demonstrates practical utility beyond academic benchmarks

4.2 Limitations and Future Work

- **Multi-person Scenes:** Currently limited to single-person videos
- **Scene Awareness:** No interaction with environmental objects
- **Long-term Dependencies:** Limited to 5-second sequences (150 frames)
- **Evaluation Metrics:** FID and diversity metrics may not fully capture perceptual quality